

Nathaniel Martes

US Citizen | nathaniel.martes.business@gmail.com | (570)-880-8723 | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

B.S. in Computer Science, Wilkes University, GPA: 3.95/4.00, Honors

Expected Graduation – May 2027

EXPERIENCE

Software Engineer Intern, NVIDIA

Incoming May 2026

- NVIDIA DGX Cloud Data Storage for AI Training and Workflows.

Software Developer Intern, VetsEZ

Sep 2025 – Current

- Built a microservice with a team of developers using Java, Spring Boot, OData and Postman, which created a REST API for legacy SQL schemas that supported crucial platform modernization efforts.
- Integrated test cases into a GitHub Actions CI/CD pipeline for a SwaggerUI OpenAPI documentation generator using Java, successfully increasing generator code coverage to 91%.
- Optimized service efficiency by building a dedicated metadata caching layer using Java, Spring Boot, Docker, AWS ECS, and PostgreSQL, reducing startup latency for dependent APIs by 3 minutes.
- Cooperated with several developers using Microsoft Teams and Github Issues, improving code quality and coherence.

System Administrator Intern, Wilkes University

Jun 2024 – Current

- Automated server administration tasks using Ansible and Linux commands, saving 1 hour during software updates.
- Received mentorship from senior system administrators during on-boarding to understand server design and tools.
- Configured Python toolchains on workstations using Ansible, making it easier for students to debug programs.
- Migrated LDAP database to LDAPS using Bash, Ansible, and Linux, ensuring heightened security for user credentials.

DevSecOps Engineer Intern, Software Engineering Institute at Carnegie Mellon University

May 2025 – Aug 2025

- Used Python, Git, Docker, and a REST API to create a vulnerability detector for installed software packages across all internally managed systems, cutting potential vulnerability detection time from 30 days down to 1 day.
- Corrected 4,400+ SQLite entries with Python, achieving 100% accuracy in quarterly software license usage reports.
- Built a Python and REST API tool to detect duplicate software license usage, saving 2 hours of manual data analysis.
- Integrated 80+ test cases for internal Atlassian suite applications into DevOps CI/CD pipelines using Python, Pytest, and REST APIs, preventing deployment of misconfigured Docker containers.
- Collaborated in an Agile environment using Jira to track project progress and participate in daily Stand-Up meetings, enhancing team communication and improving overall project understanding.

DevOps Engineer Intern, Wilkes University

Jun 2024 – Aug 2024

- Deployed a 15+ page dynamic web application using Kubernetes (K3s), Docker, Wordpress, MySQL, HTML and CSS to enhance the presentation of undergraduate research.
- Architected a K3s cluster consisting of 4 Linux nodes, to simulate an AWS Elastic Kubernetes Service (EKS) development environment, integrating the Ingress service for scalable load balancing.
- Met with professor bi-weekly to discuss server and website architecture which led to optimized design choices.

PROJECTS

Parking Lot Volume Tracking Web Application

Sep 2025 – Nov 2025

- Used TypeScript, Java, PostgreSQL, Docker, and Spring Boot to develop a full stack web application to track volumes of parking lots, creating a way for users to travel smarter and easier.
- Developed a mobile-friendly frontend interface for the app using NextJS, React, Tailwind CSS, HTML, and Vercel.
- Created GitHub Actions CI/CD pipelines and AWS CloudFormation templates to automate microservice deployments.
- Implemented a machine learning (ML) model on an edge device using Python, PyTorch, and multithreading to correctly determine when vehicles enter and exit parking lots.

AWS Affiliated Hackathon 1st Place, Interactive AI Reflection Therapy Web Application

Oct 2025 – Oct 2025

- Created a serverless full stack application using JavaScript, React, HTML, Tailwind CSS, Python, Git, and DynamoDB that creates AI interactive narrative chapters for therapy and user resonance which placed 1st among 200+ participants.
- Utilized AWS Cloud Development Kit (CDK) to create infrastructure-as-code (IoC) which deployed a REST API Gateway and Lambda functions that interacted with a NoSQL database and a Large Language Model (LLM).
- Collaborated with 2 developers to design, develop, and deploy the complete application within a 24-hour period.

Multiplayer Real Time Dungeon Crawler

Feb 2025 – May 2025

- Utilized Python, SQLite, Pygame, and FastAPI to create a full stack multiplayer dungeon crawler game.
- Designed microservice architecture which featured a REST API and Websockets for connection between players.
- Deployed CI/CD pipelines with GitLab, Docker, and Python which performed static analysis of code and ran test cases.
- Worked in an Agile environment with 2 other developers, holding bi-weekly Scrum meetings to discuss sprint plans.

J.P. Morgan Chase Affiliated Hackathon 1st Place, AI Image Disaster Analyzer

Mar 2025 – Mar 2025

- Worked in a team of 4 to develop a full stack AI image analysis application for determining room disaster readiness using Python, Javascript, TypeScript, React, HTML, and CSS which placed 1st among 470+ participants.
- Built a REST API using FastAPI and Google Gemini to perform server side prompting and AI parsing.
- Established Git and Docker workflows which lead to better team efficiency under limited development time.

LEADERSHIP

Treasurer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter

Dec 2025 – Current

- Gave proposal for national convention funding to 15+ student government members, resulting in a full funding amount of \$3800+.
- Kept track of funds from events and fundraisers using Google Sheets, ensuring organization at every step.

Vice President, Mathematics Physics and Computer Science Club at Wilkes University

Jan 2025 – Current

- Managed hackathons for attendees to hackathons in University of Delaware and Leigh University, achieving a 1st place hackathon win and university recognition.
- Created general announcements using Discord for club information, meetings, and university events for over 60+ members.

Martial Arts Instructor, Dual Dragon Karate

May 2022 – Current

- Led hundreds of classes teaching self-defense techniques and traditional forms to 15+ students.
- Handled different student age groups from 4 years old to 50+, ensuring students are satisfied with the class and that they are learning real world skills.

Public Relations Officer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter

Jan 2025 – Dec 2025

- Managed LinkedIn social media page for Wilkes Chapter, displaying student success, community involvement, general updates to over 850+ users.

SKILLS

Programming: Python, TypeScript, JavaScript, Golang, SQL

Tools: React, Spring Boot, FastAPI, HTML, CSS, Git, Docker, Linux, CI/CD, AWS, PostgreSQL, Ansible, Kubernetes